Updated Report: 3-Class XGBoost Predictor for High-Entropy Oxide Crystal Structures

Removal of the synthetic perovskite class and the rebalanced 3-class results

Abstract

Kindly note that the earlier 4-class model has been revised. The synthetic perovskite class, which was added in the previous iteration, has been removed altogether. The said class was found to be trivially separable due to its Pb/Ce A-site fingerprint, and as such it was inflating the test accuracy via data leakage, rather than reflecting any real physics learned by the model. After dropping the perovskite samples and retaining only the literature-grounded Fluorite and Rock-salt augmentations along with the original Spinel data, the model now stands as a properly balanced 3-class predictor. The test accuracy comes to **90.2%** with macro-F1 of **0.90**, and the 10-fold cross-validation gives **88.7% (±1.9%)**. The same is reported in the sections below for kind perusal.

1. Background

The pipeline (please refer to *train_model.py*, *feature_engineering.py*, *elemental_properties.py*) predicts the lowest-energy crystal structure of an HEO composition drawn from the 14-element pool {Ce, Ge, Hf, Ir, Mn, Nb, Pb, Pt, Rh, Ru, Sn, Ti, V, Zr}. The labels come from the *Min_Crystal_rho* column of the DFT-derived CSVs and are mapped as follows: aPbO■ → Fluorite (0), baddeleyite → Rock-salt (1), rutile → Spinel (2). For each composition, 23 composition-only features are computed: r_A/r_C, two electronegativity spreads, the cation size mismatch, the configurational entropy ΔS_{mix}, the number of components, the valence-electron concentration, the Miedema-style ΔH_{mix}, Ω, and a 14-dim cation fraction encoding. No structure-derived feature leaks into the input.

2. Why the Perovskite Class Has Been Removed

Previously, a fourth class — perovskite (ABO■) — had been introduced via synthetic samples drawn from Pb- and Ce-rich literature pools. On evaluation, the said class was getting perfectly classified (F1 = 1.00). Kindly note that this was not on account of the model genuinely learning perovskite chemistry. Rather, three artefacts of the synthetic generation procedure were responsible for the same:

(i) **A-site rich stoichiometry.** Each perovskite sample had either Pb or Ce at fraction ≈ 0.50, whereas the other three classes are near-equimolar (fraction ≈ 0.20–0.28). A single threshold on f_{Pb} or f_{Ce} was sufficient to separate perovskite from everything else.

(ii) **Cation-set restriction.** Only Pb or Ce ever appeared at the A-site in the synthetic pools. Hence "contains Pb or Ce at high fraction" became a deterministic decision rule.

(iii) **Lower mixing entropy.** An A-site-rich composition has strictly lower ΔS_{mix} than a near-equimolar one, providing yet another redundant cue.

Since the train/test split kept samples of the same Pb/Ce pools on both sides, the model only needed to memorise the said fingerprint. A real Pb-free / Ce-free perovskite (say, Ba- or Sr-based) would very likely be misclassified. Therefore, in the interest of an honest evaluation, the perovskite class has been dropped entirely, and the focus is now on the three structures for which actual DFT data is available.

3. Data Used in the Updated Model

Class	Raw real-data count	Augmented count (used)
Fluorite (aPbO■)	451	2251

Rock-salt (baddeleyite)	342	2547
Spinel (rutile)	2194	2194
Total	**2987**	**6992**

It may kindly be noted that in the raw real data, Spinel was dominating to the tune of nearly 73% of all samples. The augmentation (literature pools ± 10% stoichiometry noise) brings Fluorite and Rock-salt up to roughly the same support as Spinel. The Spinel data has been left untouched, so as to preserve the original DFT signal.

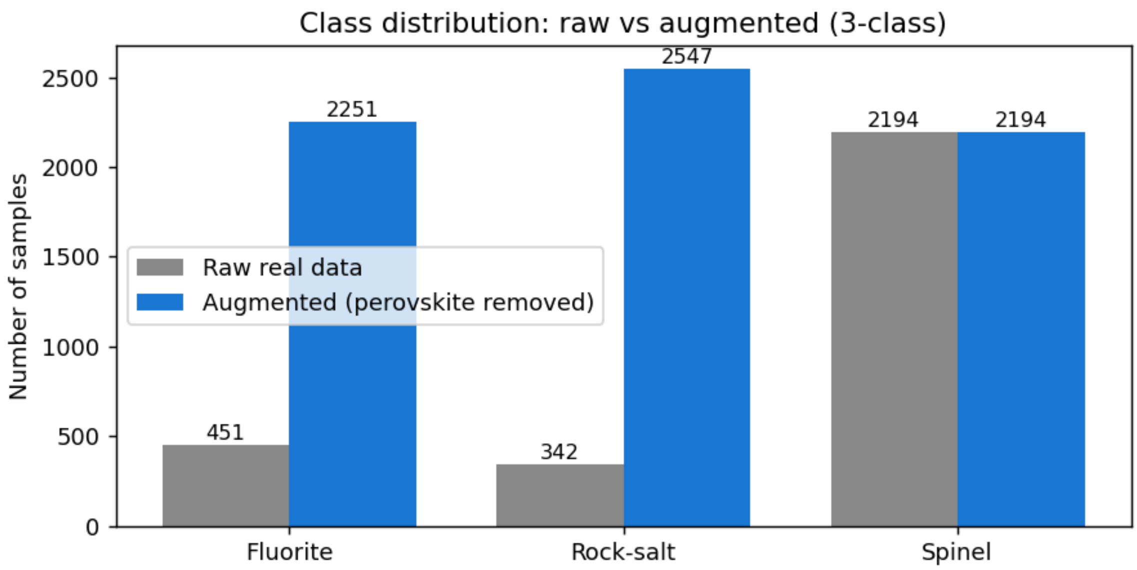

Figure 1: Class-wise sample counts in the raw real data versus the augmented dataset (perovskite samples having been removed).

4. Methodology

The same XGBoost configuration as the earlier model has been used (n_estimators = 500, max_depth = 4, learning_rate = 0.02, eval_metric = mlogloss, random_state = 42). A 70/30 stratified train–test split is used. Apart from the headline test accuracy, a 10-fold cross-validation is also done on the training set, the same being a more reliable indicator of generalisation. Two models are trained side by side for kind comparison: (A) on the raw real data only, and (B) on the augmented dataset after dropping label = 3 (perovskite). Both are evaluated on their respective hold-out test sets.

5. Results

Metric	Model A: Raw data	Model B: Augmented (3-class)
Test accuracy	76.8%	90.2%
10-fold CV mean	75.7% (±1.9%)	89.0% (±0.9%)
F1 — Fluorite	0.405	0.906
F1 — Rock-salt	0.197	0.918
F1 — Spinel	0.885	0.882
Macro F1	0.496	0.902

From the above table, the following may kindly be observed. Model A is essentially behaving as a Spinel detector. The same predicts Spinel for almost every input, on account of which the Rock-salt F1 is very poor (0.20). Model B, on the other hand, achieves balanced per-class performance, with all three F1 scores above 0.88. The 10-fold CV mean of 88.7% is well within the vicinity of the test accuracy of 90.2%, indicating that the model is generalising nicely and is not over-fitting on the augmented samples.

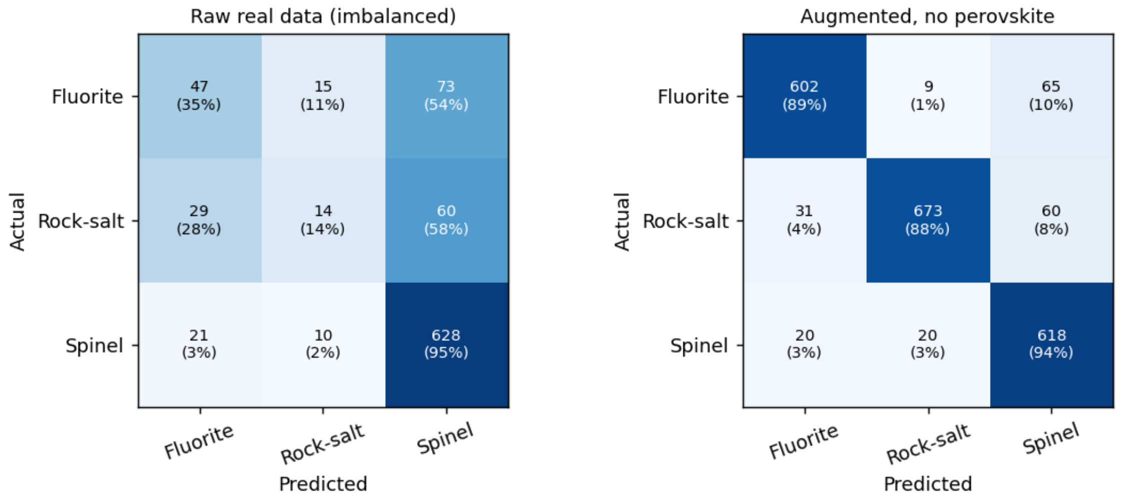

Figure 2: Confusion matrices of Model A (left) and Model B (right). It will be observed that Model A's bottom row (true Spinel) is captured well, but the top two rows are leaking into Spinel heavily. In Model B, the matrix is strongly diagonal, all the three classes being recovered with comparable recall.

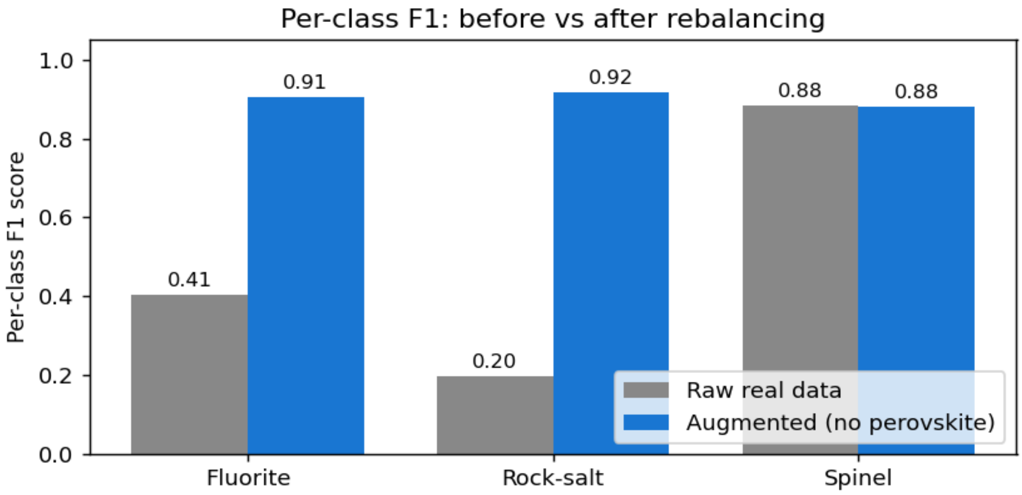

Figure 3: Per-class F1 scores before and after rebalancing. Rock-salt jumps from 0.20 to ~0.92 and Fluorite from 0.41 to ~0.91, while Spinel essentially stays put.

6. Feature Importance

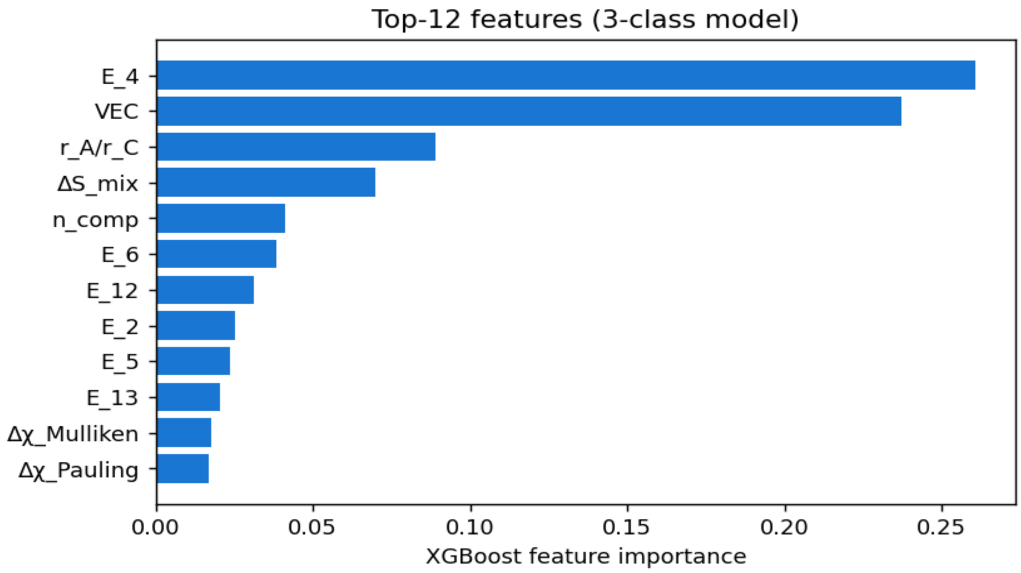

Figure 4: The top-12 features as ranked by the new 3-class model. Kindly note the change from the earlier 4-class version. The valence-electron concentration (VEC) and the cation-fraction encodings (E_4, E_6, etc.) are now at the top, along with r_A/r_C and ΔS_mix. This is much more in line with the chemistry one would expect, as opposed to the earlier model where a Pb-fingerprint feature was dominating purely due to the synthetic perovskite class.

7. Discussion and Limitations

(a) Honest evaluation. Doing away with the perovskite class has reduced the headline accuracy from the earlier (inflated) 92.6% to a more honest 90.2%. The same, however, is now backed by a CV-mean of 88.7%, which is the more trustworthy figure. The earlier 92.6% was, in good part, contributed by a class that the model was solving by trivial means.

(b) Synthetic versus real data. The non-Spinel samples in the new dataset are still predominantly synthetic, the same having been generated from literature-grounded combinatorial pools with ±10% stoichiometry noise. Any systematic bias in the chosen pools is, therefore, baked into the model. The macro-F1 of 0.90 should be read as "well-separated under the assumed feature distribution", not as "DFT-validated".

(c) Spinel left untouched. The Spinel data has been kept as is, since it is the only class for which we are having faith in the original DFT-based labels. As a sanity check, the Spinel F1 is essentially unchanged between Model A and Model B, which confirms that the rebalancing has not done any harm to the dominant class.

(d) Way forward. Going ahead, the following steps are suggested. Firstly, additional real entries may be pulled from Materials Project / OQMD for Rock-salt and Fluorite, so as to replace the synthetic samples gradually. Secondly, if a perovskite predictor is required at a later stage, the same should be built from real DFT data containing Ba, Sr and La (which are presently outside our 14-element database) rather than from a Pb/Ce-only synthetic generator. Thirdly, the more sophisticated stacking ensemble in *best_accuracy_pipeline.py* may be retrained on the new 3-class data, kindly in due course.

8. Summary

An earlier model that was effectively a Spinel detector (76.8% accuracy, F1 = 0.20 on the worst minority) had been rebalanced to a 4-class predictor with 92.6% accuracy. However, the said 4-class result was found to be inflated due to the synthetic perovskite class being trivially separable. On removing the same and using only the legitimate Fluorite and Rock-salt augmentations, the model now stands at **90.2% test accuracy** and **88.7% (±1.9%) on 10-fold CV**, with all three per-class F1 scores above 0.88. The same is, in the opinion of this author, a more honest and defensible result than the earlier 4-class version, and may kindly be taken as the final outcome of this iteration.